

Penetrating Wounds of the Neck

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OF all the topographic regions associated with injury in man, the area of the neck is most vulnerable to multiple anatomic and physiologic derangement. Injury from a bullet may result in a small penetrating cervical wound and still have traversed the trachea, the esophagus and a major blood vessel with few immediate clinical signs. Although the relative incidence of penetrating wounds of the neck is as frequent in civilian life as on the battlefield, diverse views regarding the need for early surgical exploration still prevail. The present study was undertaken in an effort to evaluate the need for early surgical exploration of all penetrating neck wounds as well as to correlate results obtained with the form of therapy employed.

The majority of reports dealing with wounds of the neck have been derived from the surgery of warfare. Only three neck wounds were reported from the Crimean War. Two of these were fatal and the third resulted in invalidism.¹ During the Civil War there were 4,114 cases of wounds of the neck with an approximate mortality of 15 per cent.² The relatively low mortality rate in that series may be attributed to the management of neck wounds during that period. The dangers of secondary hemorrhage resulting from the expectant therapy of vascular injury were well described. The importance of an adequate airway, wound débridement and wound drainage was a considered aspect of therapy.

From the Spanish American War 188 neck wounds were reported with a mortality of 18 per cent.³ During World War I there were 594 cases reported with an approximate mortality of 11 per cent.⁴ However, the great number of arterial aneurysms in the neck reported as delayed complications of neck wounds was evidence that expectant therapy was employed too often. The greatest number of deaths resulted from uncontrolled hemorrhage.

In the period between the two wars the litera-

ture was characterized by single case reports.⁵⁻⁸ It is difficult to state what the general mortality in civilian neck wounds was during this period. An increasing number of arteriovenous aneurysms in the neck were reported and this fact may be interpreted as indicative of an increasing incidence of delayed exploration of neck wounds for vascular injury. Although great refinements in general surgical care were made up to the time of World War II, the advancements in the treatment of shock, infection and the metabolic aspects of injury were not well correlated with the technical aspects of penetrating neck injury. Although the errors of inadequate and delayed surgery had been well demonstrated in earlier years, sporadic reports of cases presenting the sequelae of surgical delay continued to appear.⁹⁻¹¹

World War II afforded an impetus to more definitive expression of the surgical management of neck wounds. Tracheotomy appeared to have been employed at an earlier period; adequate drainage and control of infection for pharyngoesophageal wounds was instituted; and early therapy of vascular injury with attempts to restore continuity of the vessel was advocated. One series reported 851 cases of wounds of the neck with a mortality of 7 per cent.^{12a} Lichtenstein^{12b} reported a series of cases emphasizing the importance of anesthesia, early operation, adequate débridement and drainage. The need for early exploration of every cervical hematoma was strongly advised by Bailey.¹³

While combined organ injury in the neck does not permit standardization of surgical therapy, the importance of early exploratory cervicotomy cannot be overemphasized. Despite the advocacy over 100 years ago of exploration of all wounds penetrating the platysma, bleeding neck wounds are still occasionally treated with pressure, expectant therapy and observation. Tracheal and esophageal injury occasionally is undiagnosed and results in death or serious complications. The following series of cases admitted to one county hospital is presented with the hope that it may